

End-to-End AQI Prediction System

A Serverless AQI Forecasting System for Islamabad

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1. Introduction

This project implements a **100% serverless AQI prediction system for Islamabad, Pakistan**.

The system automatically gathers data on weather and air quality, develops useful features from it, stores these features in the Hopsworks Feature Store, trains a number of machine-learning models, registers the best of these models, and then predicts the AQI for the next **72 hours**.

GitHub Actions is used to automate the hourly ingestion of data and the daily retraining of models, whereas Streamlit offers an interactive dashboard for forecasts, model evaluation, explainability, EDA, and health alerts.

2. System Architecture and Pipeline Design

The system is based on a serverless, data-centric machine learning architecture and combines automated data ingestion, feature engineering, model training, model versioning, inference, and visualization through the use of GitHub Actions, Hopsworks, and Streamlit.

Stage	Description	Technology
Data Ingestion	Performs historical backfill once and collects current weather and pollutant data every hour	GitHub Actions, Open-Meteo
Feature Engineering & Storage	Creates temporal, seasonal, wind, weather interaction, and other derived features, then stores them in the Feature Store	Python, Hopsworks Feature Store
Model Training	Retrieves historical features, creates AQI lag and rolling features, trains multiple regression models, and evaluates them using a chronological split	Scikit-learn, XGBoost
Model Evaluation & Explainability	Compares models using RMSE, MAE, and R^2 , selects the best model, and generates SHAP feature importance	Scikit-learn, SHAP

Model Registry	Stores and versions the selected model together with metrics, feature information, and SHAP artifacts	Hopsworks Model Registry
Inference	Fetches future weather and pollutant forecasts and generates recursive AQI predictions for the next 72 hours	Python, Open-Meteo, Hopsworks
Dashboard	Displays current AQI, 3-day forecast, model performance, SHAP explainability, EDA, and health advisories	Streamlit, Plotly
CI/CD Automation	Runs hourly feature ingestion and daily model retraining automatically	GitHub Actions

Figure 1: System Component Overview

3. Data and Feature Pipeline

The project uses Open-Meteo for weather and air-quality data.

Main inputs include:

- Temperature
- Relative humidity
- Wind speed and direction
- PM2.5
- PM10
- CO
- NO₂
- SO₂
- Ozone
- US AQI

Approximately **90 days of historical hourly data** were backfilled initially.

The feature pipeline creates:

- Hour, day, month and seasonal features
- Weekend and rush-hour indicators

- Wind U/V components
- Stagnation indicator
- Temperature-humidity interaction
- AQI lag features
- AQI rolling averages

The engineered features are stored in the **Hopsworks Feature Store**.

The feature pipeline runs automatically every hour using GitHub Actions.

4. Model Training and Evaluation

Four regression models were trained:

1. Ridge Regression
2. Random Forest
3. Gradient Boosting
4. XGBoost

A chronological **80% training / 20% testing split** was used.

The models were evaluated using RMSE, MAE and R^2 .

Model	RMSE	MAE	R^2
Ridge Regression	6.482	4.627	0.931
Random Forest	5.745	3.839	0.946
Gradient Boosting	5.330	3.597	0.953
XGBoost	5.819	4.074	0.945

The best-performing model was **Gradient Boosting** with:

- **$R^2 = 0.953$**
- **RMSE = 5.33**
- **MAE = 3.60**

The model that has been selected is kept and versioned in the Hopsworks Model Registry.

The training pipeline runs automatically every day.

5. Forecasting and Explainability

The system generates hourly AQI predictions for the next **72 hours**.

Because future AQI values are unavailable, the model uses **recursive forecasting**:

Known AQI History



Predict Next Hour



Use Prediction as Lag



Predict Following Hour



Continue for 72 Hours

SHAP is used to explain which features have the greatest influence on model predictions.

6. EDA and Dashboard

Exploratory Data Analysis is included in the Streamlit dashboard.

The dashboard provides:

- Historical AQI trend
- AQI distribution
- Average AQI by hour
- Average AQI by weekday
- Pollutant correlation with AQI
- SHAP feature importance
- Model performance metrics

The dashboard also shows:

- Current Islamabad AQI
- PM2.5, PM10, NO₂ and Ozone
- Three-day AQI forecast
- 72-hour AQI trend
- Daily minimum, maximum and average AQI
- Health advisory
- Detailed hourly forecast

Current AQI comes from **Open-Meteo/CAMS**, while the next 72-hour AQI values are generated by the trained machine-learning model.

7. Automation

The project is fully automated using GitHub Actions.

Hourly Feature Pipeline

- Fetch latest data
- Engineer features
- Store latest row in Hopsworks

Daily Training Pipeline

- Retrieve historical features
- Generate lag features
- Train multiple models
- Evaluate models
- Generate SHAP information
- Register the new model

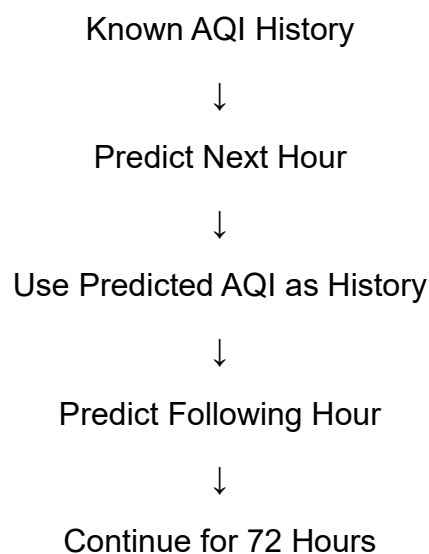
API credentials are stored securely using GitHub Secrets and Streamlit Secrets.

8. Findings and Standout Contribution

The key result of the project is that the recent history of the AQI is very useful for forecasting the AQI in the short term; lag values of the AQI and moving averages enabled the model to capture short-term pollution patterns, while information on weather and pollutants gave it further context.

Among the tested models, **Gradient Boosting performed best**, achieving an R^2 score of **0.953**, with RMSE **5.33** and MAE **3.60**.

What sets this project apart most is the method used for generating the 72-hour forecast. The future AQI values are not obtained directly from the API; rather, the model predicts them recursively.



This makes the three-day forecast a genuine machine-learning prediction rather than a direct transformation of future AQI data.

A major contribution is the full serverless MLOps pipeline, since the system automatically gathers data on an hourly basis, retains the model every day, stores the features and various versions of the model in Hopsworks, produces SHAP explanations, and serves the results using Streamlit, there being no need for a continuously running backend server.

So the main standout points are:

- **Real recursive 72-hour AQI forecasting**
- **AQI lag and rolling-history features**
- **SHAP-based explainability**
- **Hourly ingestion and daily retraining**
- **100% serverless end-to-end workflow**